

Clinical samples and metabolome coverage : choices and compromises

Justine Bertrand-Michel

5 octobre 2026

THEMATIC SCHOOL #2 • RFMF

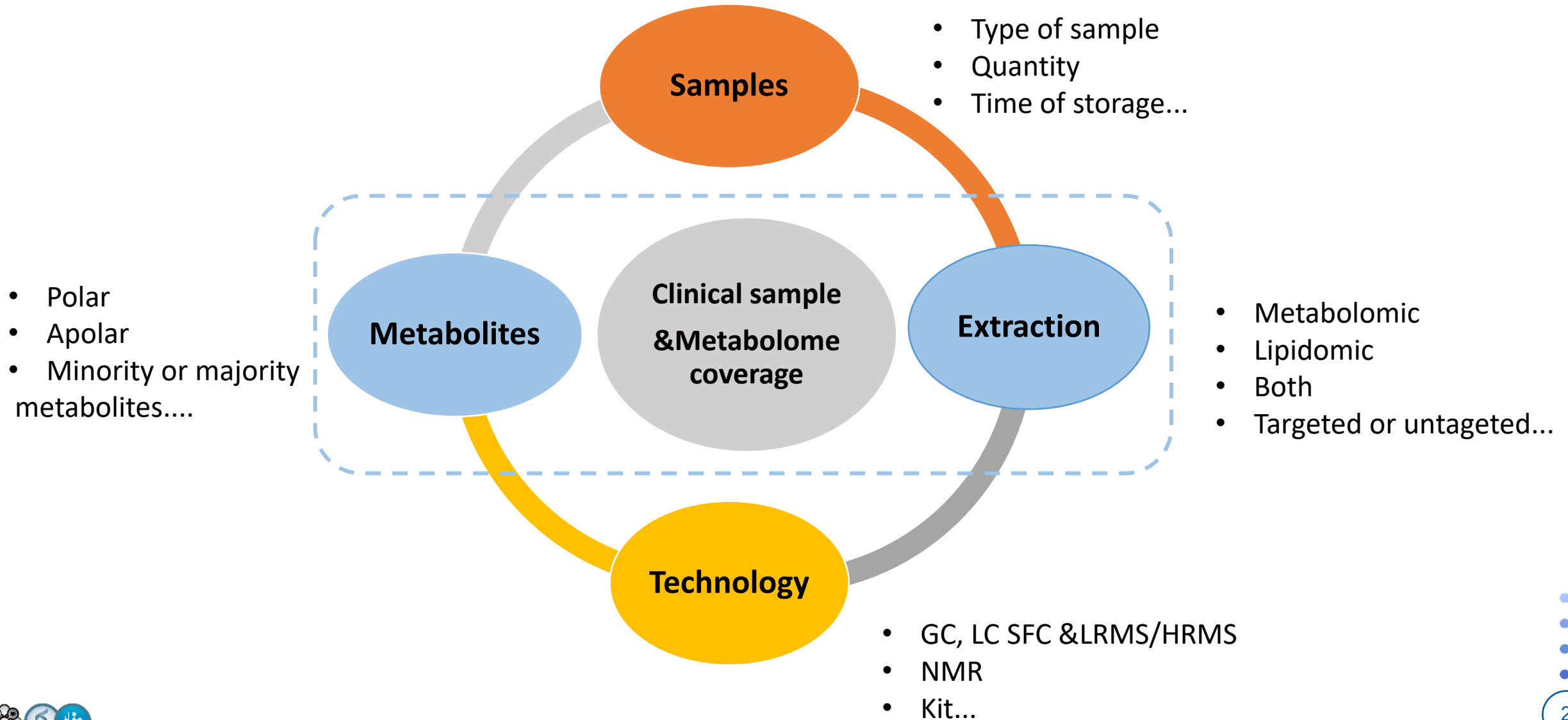
**Metabolomics
& Human Health**



October 5-9, 2026

Paul Langevin Center CAES CNRS, Aussois, Savoie

Issues



Nature of samples?

❖ Fluids



Plasma,
serum



Saliva

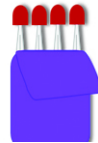


Urine

Cerebrospinal fluid...



❖ Special devices

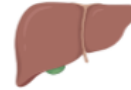


For blood: DBS, VAMS...



For skin: Tape stripping,
cigarette paper...

❖ Biopsies



Various organs...



❖ Others



Feces...



Collect expiration...



❖ Relevance of the nature of sample to the biological question

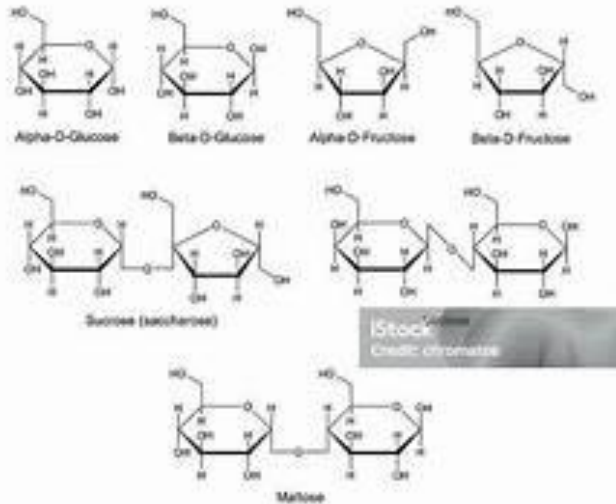
❖ Storage vs metabolite stability



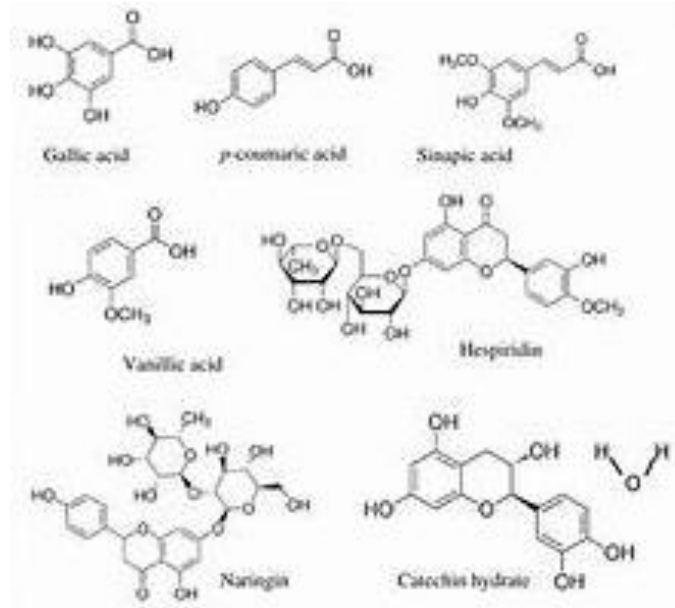
Nature of metabolites...

❖ Polar metabolites = soluble in water

Chemical structures of main sugars

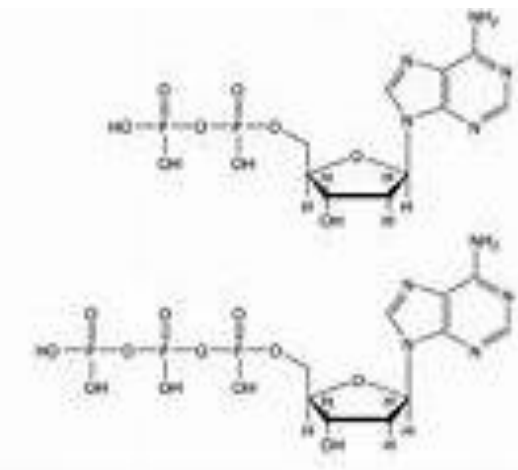
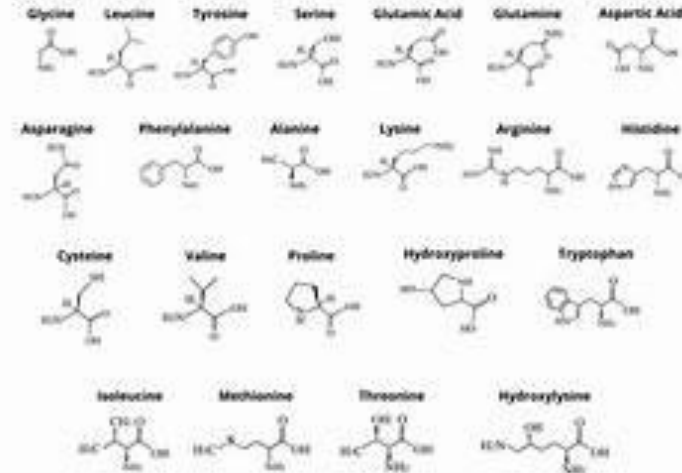


Sugar



Organic acid

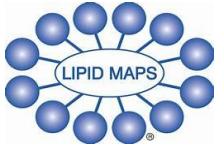
Amino Acids



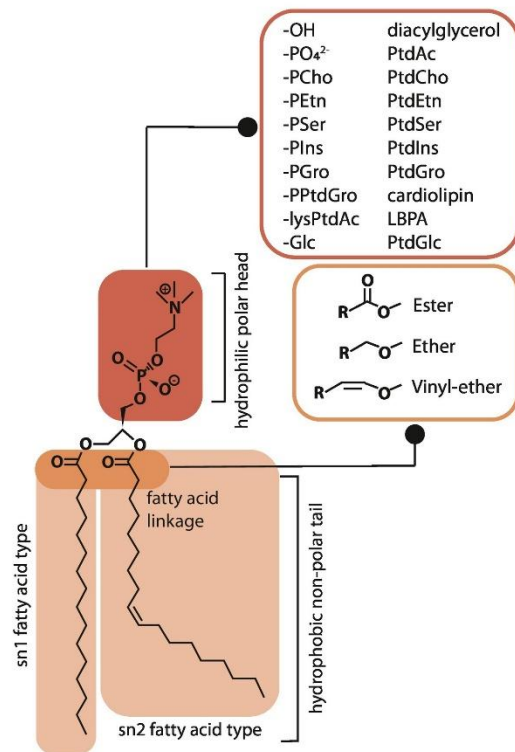
Phosphorilated nucleoside

Nature of metabolites...

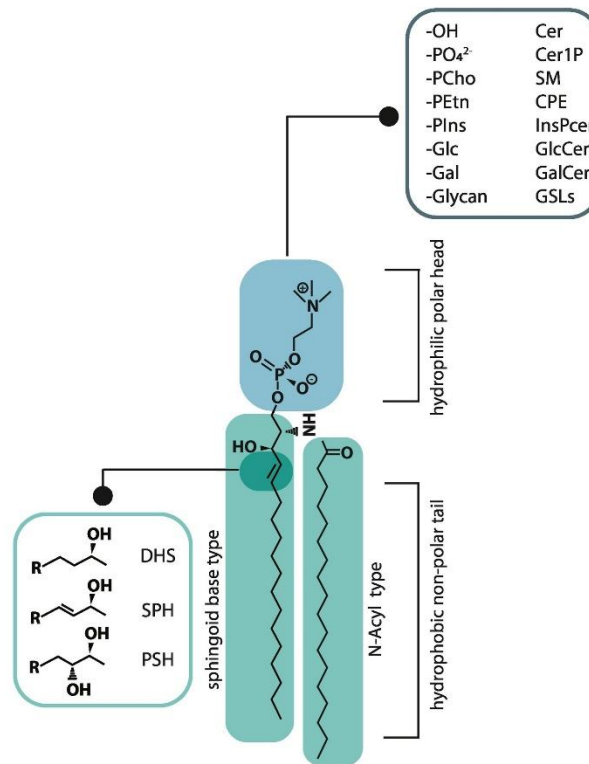
❖ Apolar metabolites = lipids = soluble in organic solvent



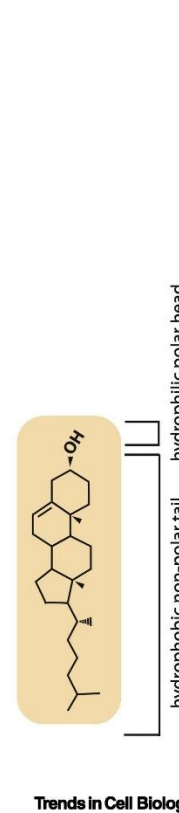
(A) Glycerolipids



(B) Sphingolipids



(C) Sterols



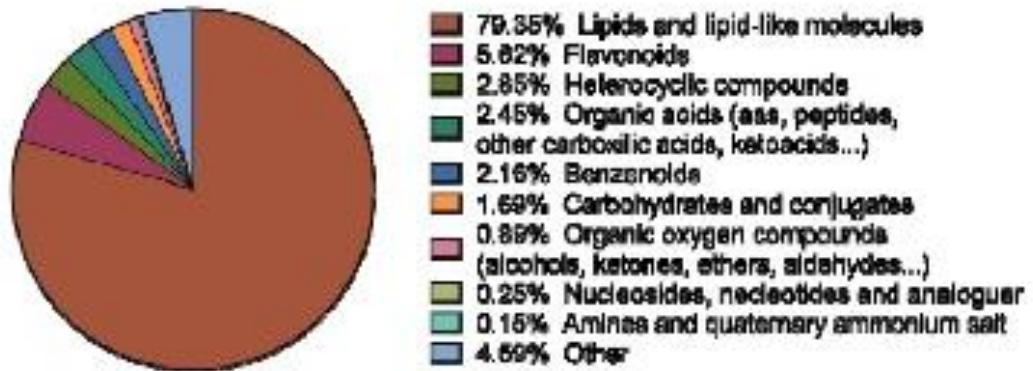
Lipid Category	Curated	Computationally-generated	All
Fatty Acyls [FA]	9736	1897	11633
Glycerolipids [GL]	411	7378	7789
Glycerophospholipids [GP]	2308	8297	10605
Sphingolipids [SP]	1844	3168	5012
Sterol Lipids [ST]	4055	0	4055
Prenol Lipids [PR]	2627	0	2627
Saccharolipids [SL]	57	1294	1351
Polyketides [PK]	7205	0	7205
TOTAL	28243	22034	50277

Number of structures per lipid category

...diversity !

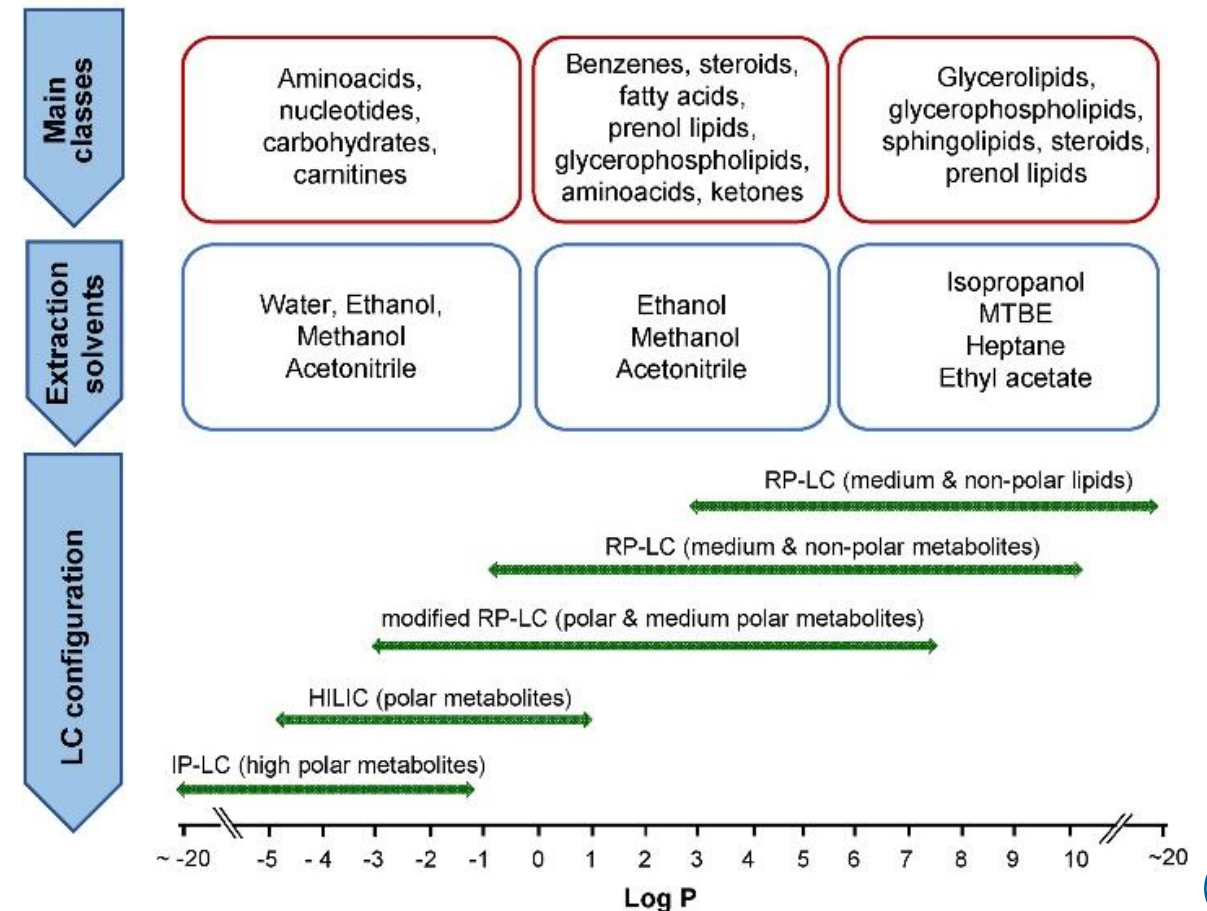
❖ High diversity of structures

Expected human metabolome
~ 118000 COMPOUNDS



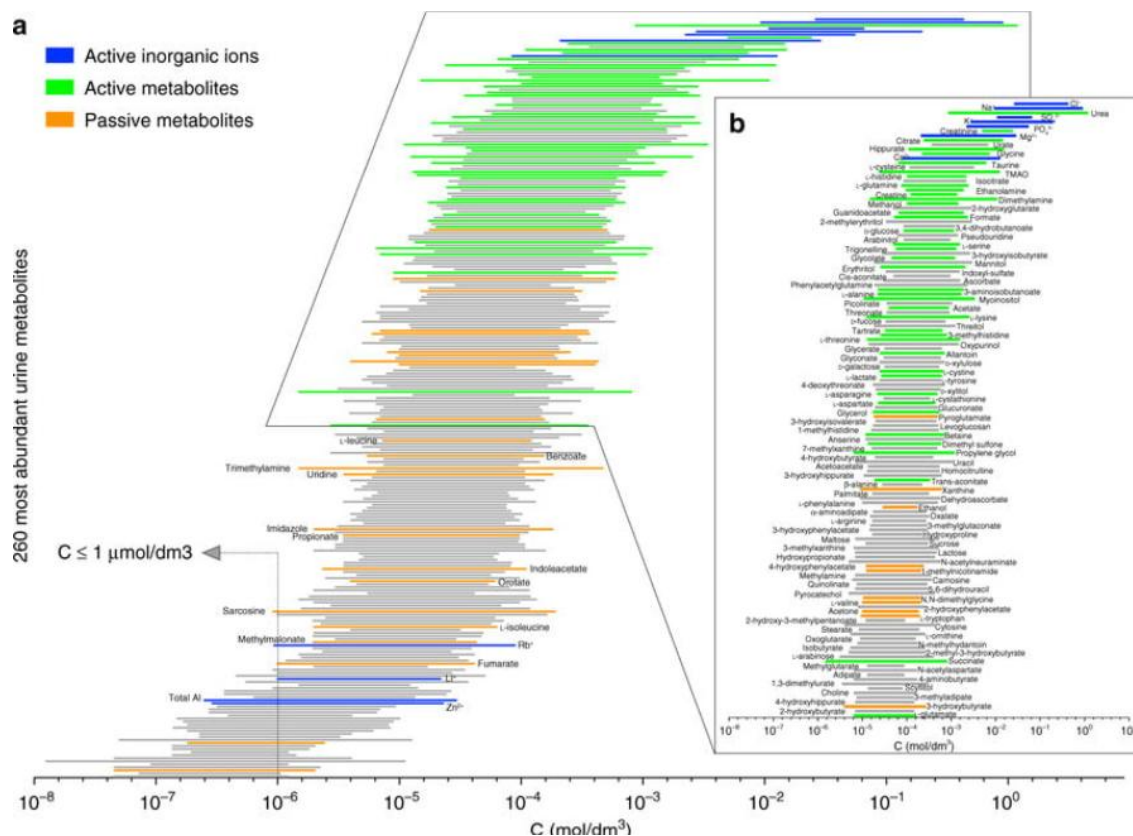
✓ Taxonomic classification (HMDB and KEGG)

❖ Large physical properties



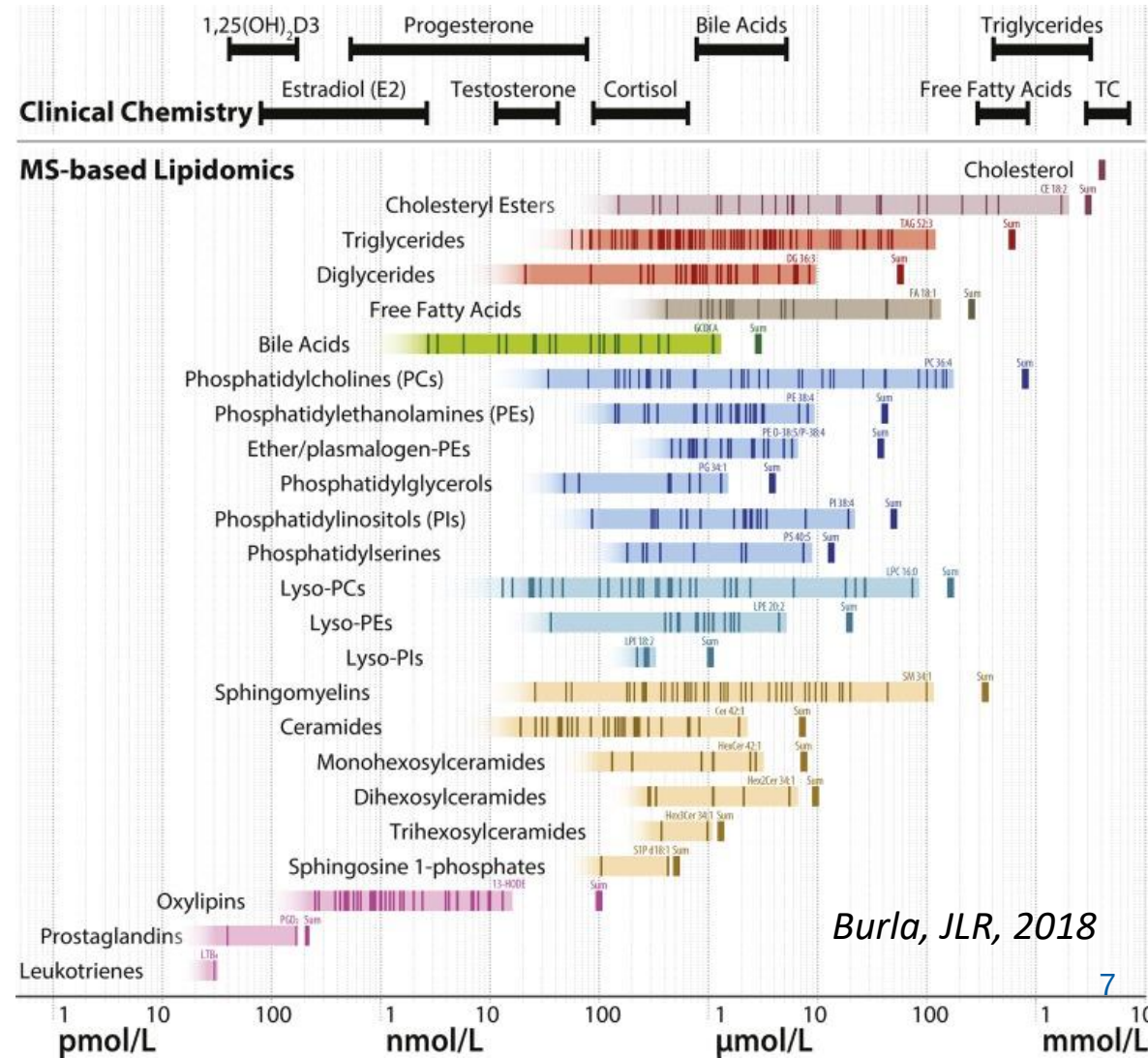
Large gamme of concentration

❖ Metabolites in human urine



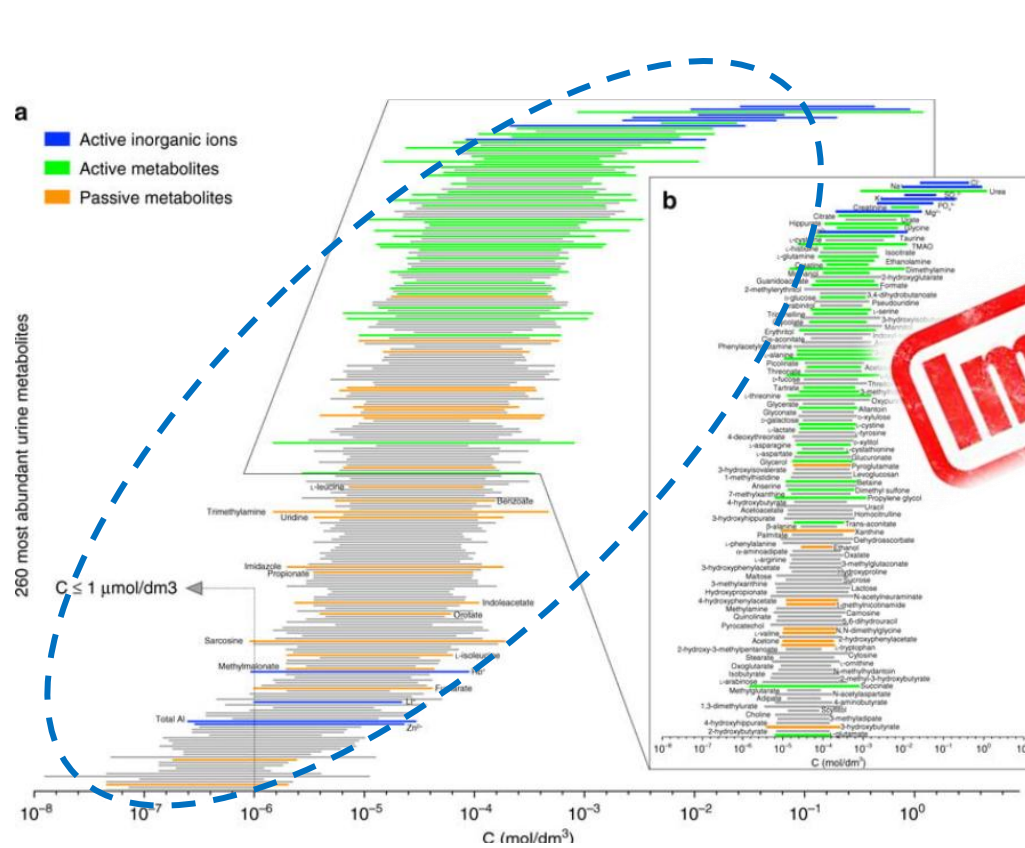
Takis, Nature, 2027

❖ Lipids in human plasma

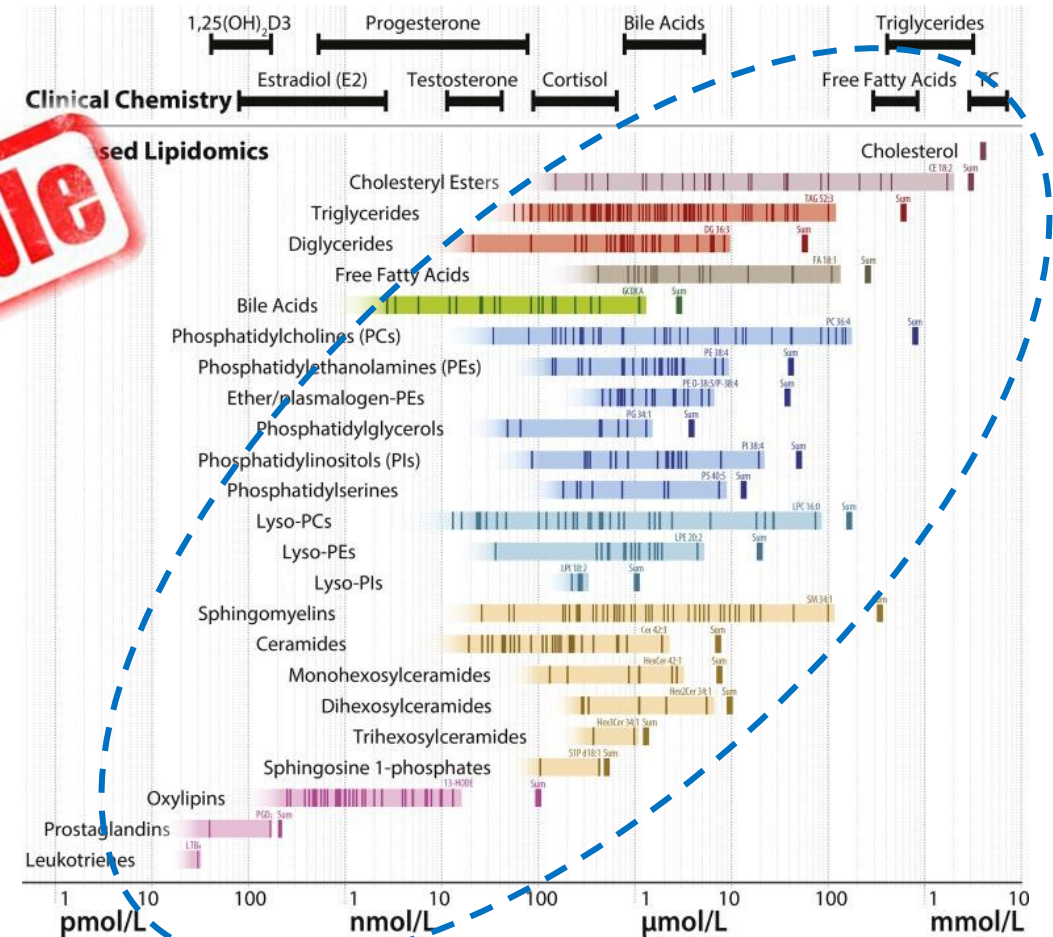


Which approaches in metabolomic?

❖ Analyse a complete metabolome/lipidome ?



Takis, Nature, 2027

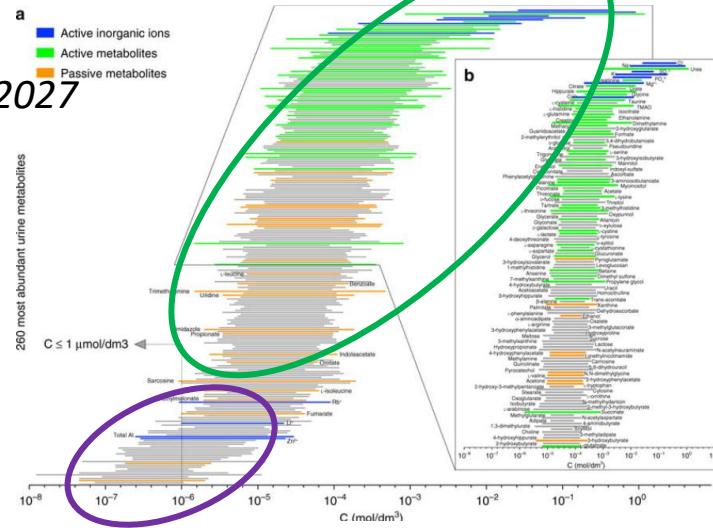


Burla, JLR, 2018

Untargeted & targeted metabolomic

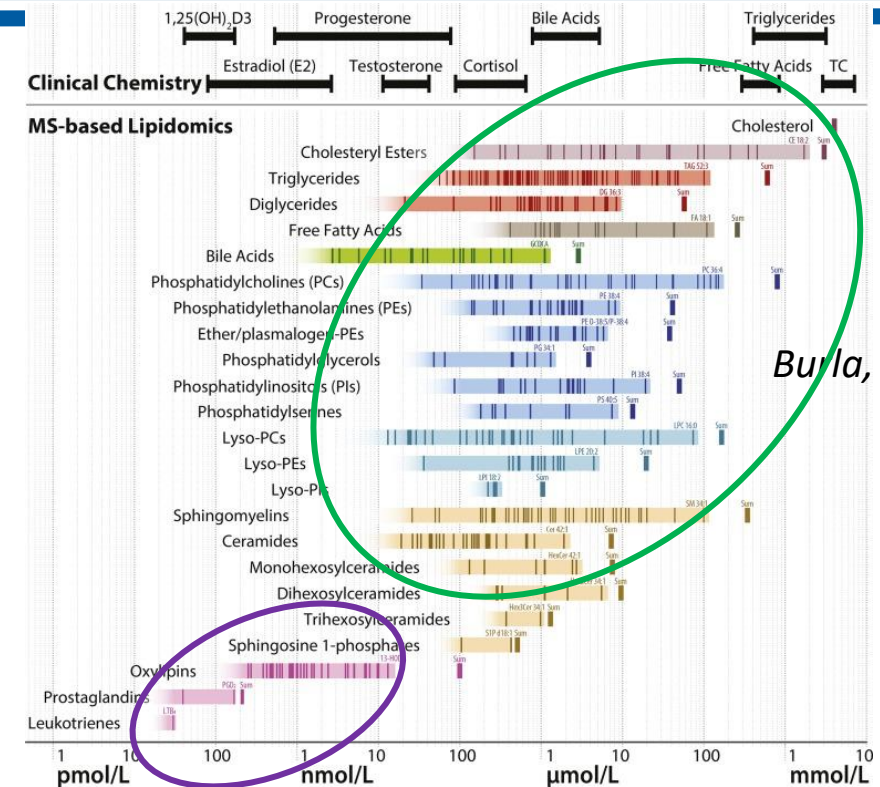
❖ Analyse a complet metabolom/lipidome ?

Takis, Nature, 2027



✓ Untargeted method

- No apriori
- Statistic → discriminant metabolite → annotation
- Most abundant metabolites
- Relative quantification



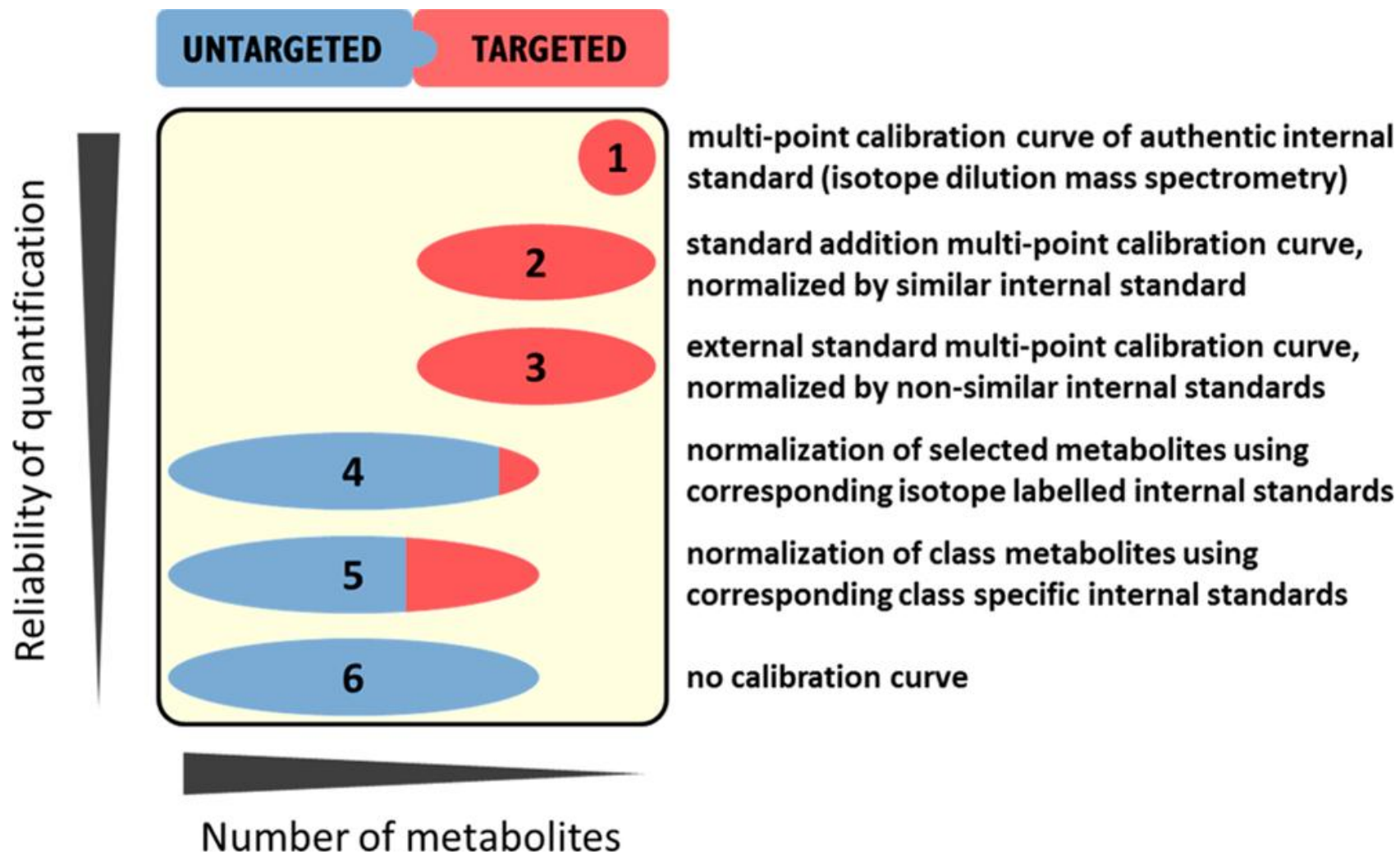
Buila, JLR, 2018

✓ Targeted method

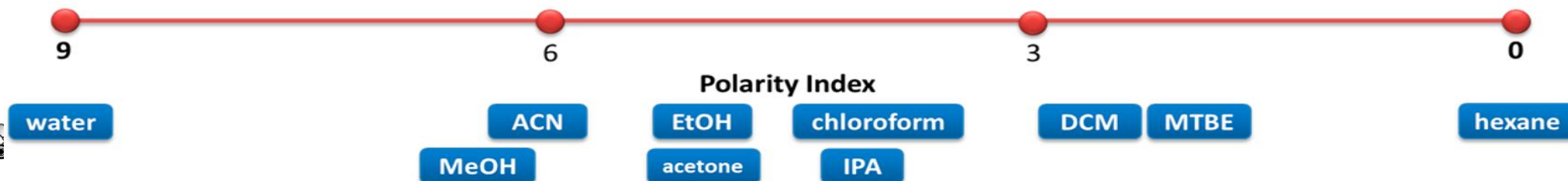
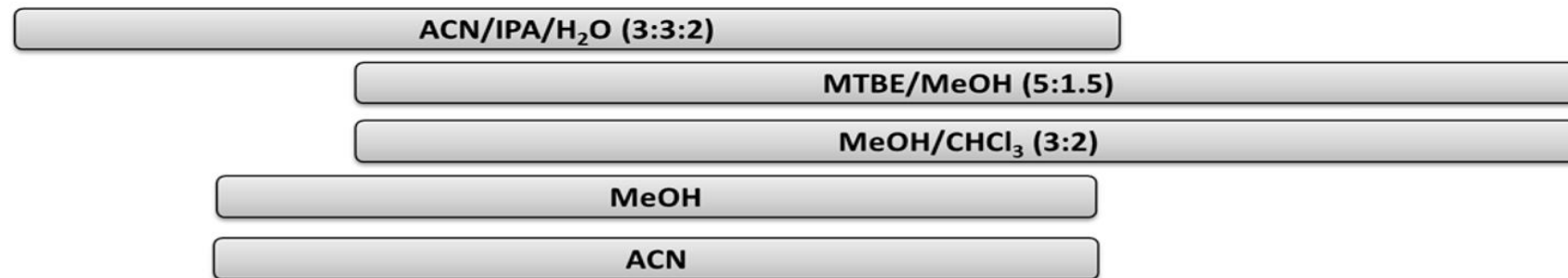
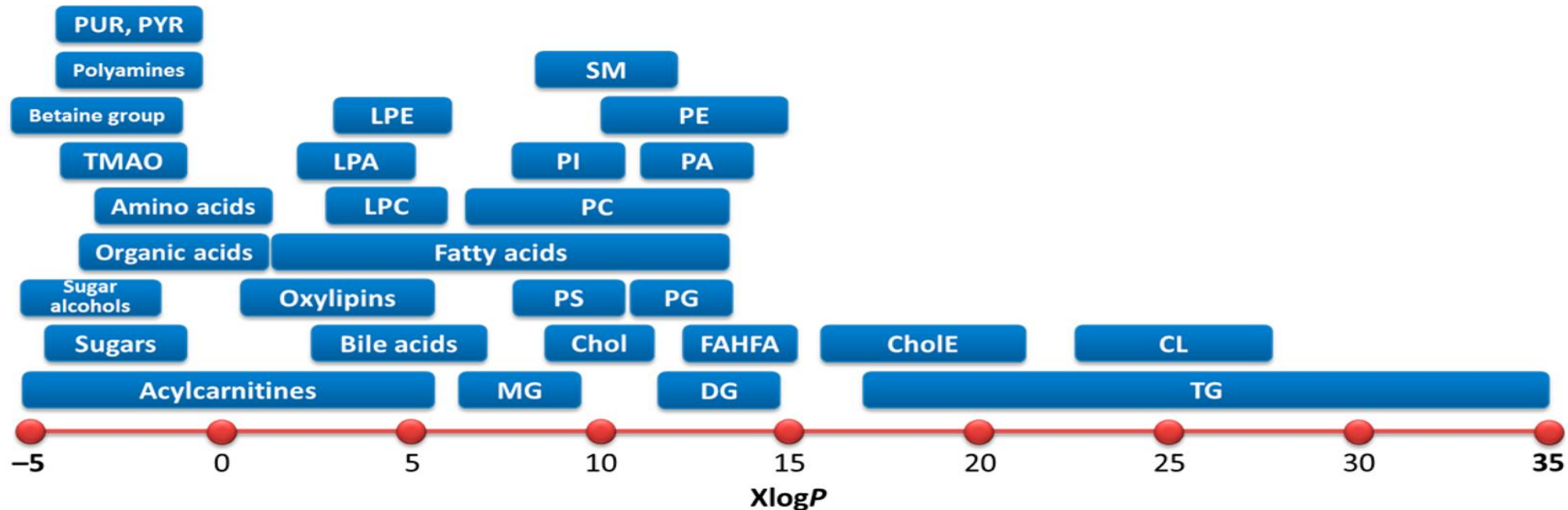
- Target a metabolite/sub-family
- Lowest abundant metabolites
- Absolute quantification



Untargeted vs targeted



Biophysical properties and extraction



Sample preparation

Samples



Sampling & Storage

- Preparation/dissection
→ Snap frost
- +/- freeze-drying
- At least -80°C
- No thaw/freezing cycle
- +/- anti-oxidant

Normalisation

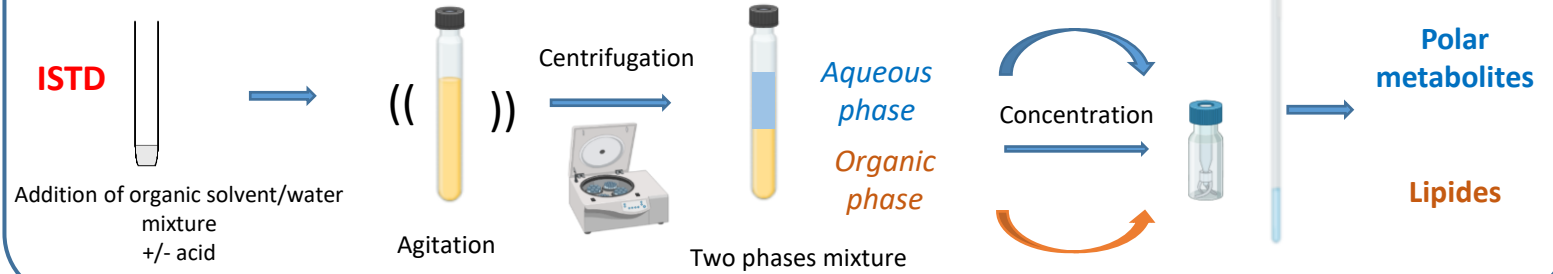
- Volume
- mg of tissue
- mg of protein
- Quantity of DNA
- Number of cells
- OD...

Crushing

- Grinding in solution
- Cold grinding...



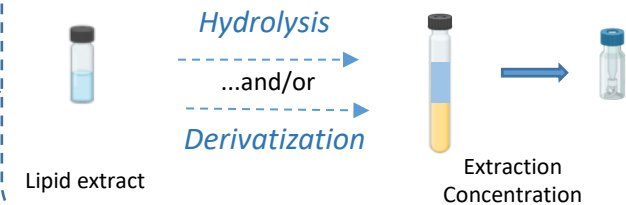
Liquid-liquid extraction



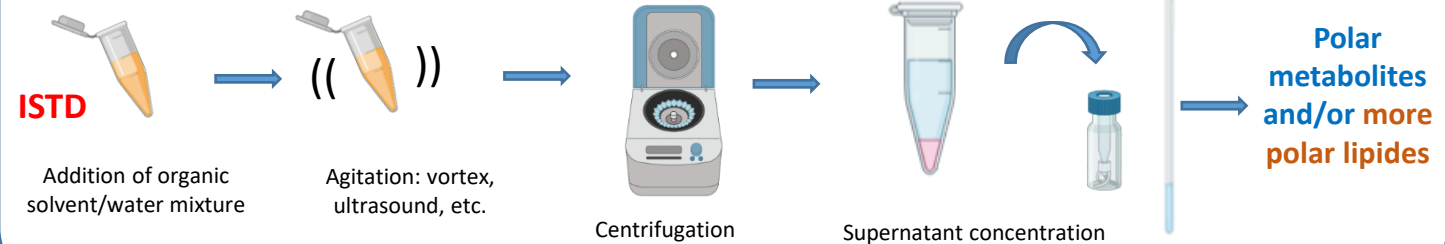
SPE extraction



Chemical modification



Protein precipitation



Sample preparation



❖ To be adapted to the biological question

❖ Number of sample

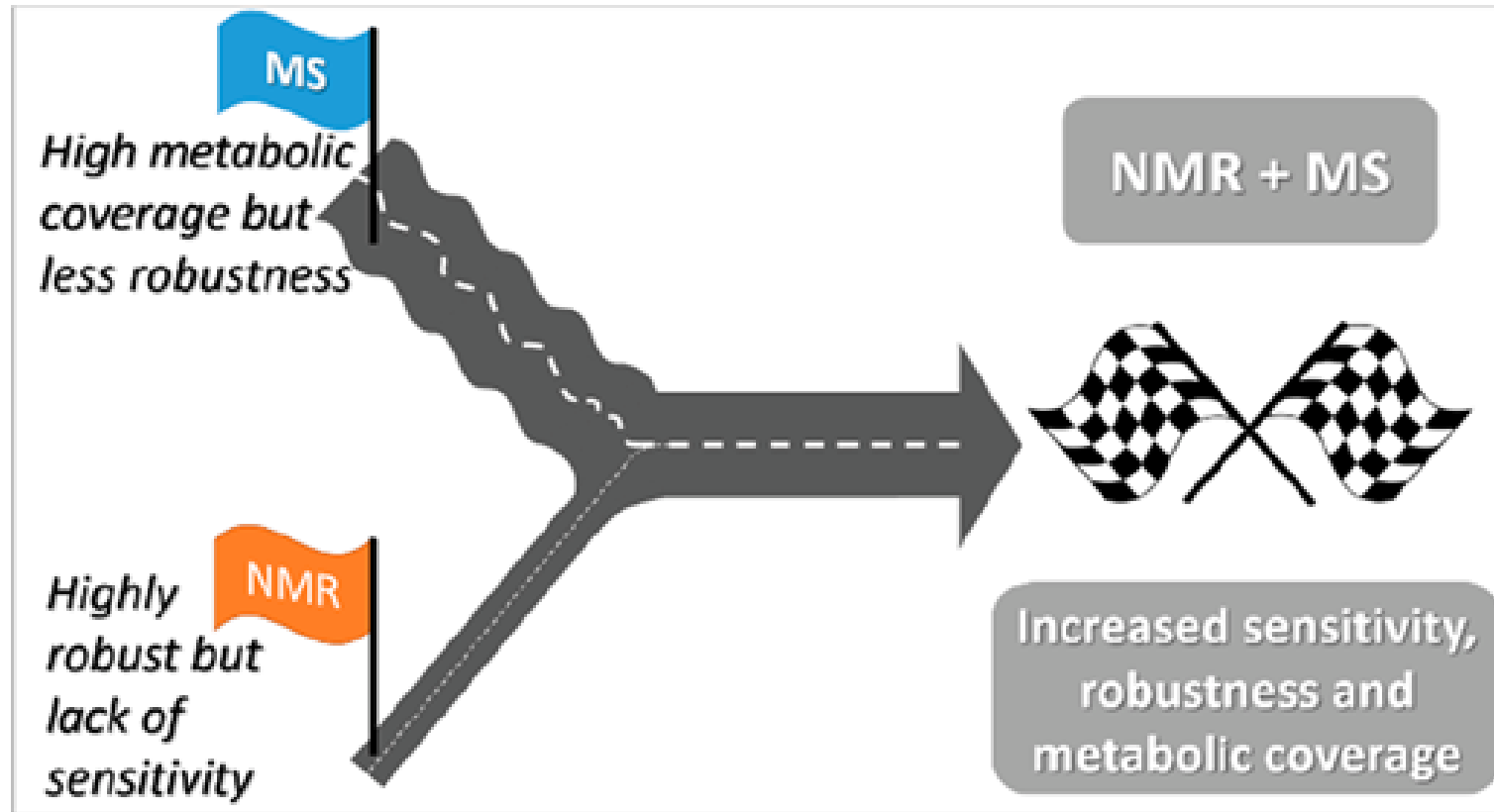
❖ Possibility of automatisatisation

❖ Single extraction

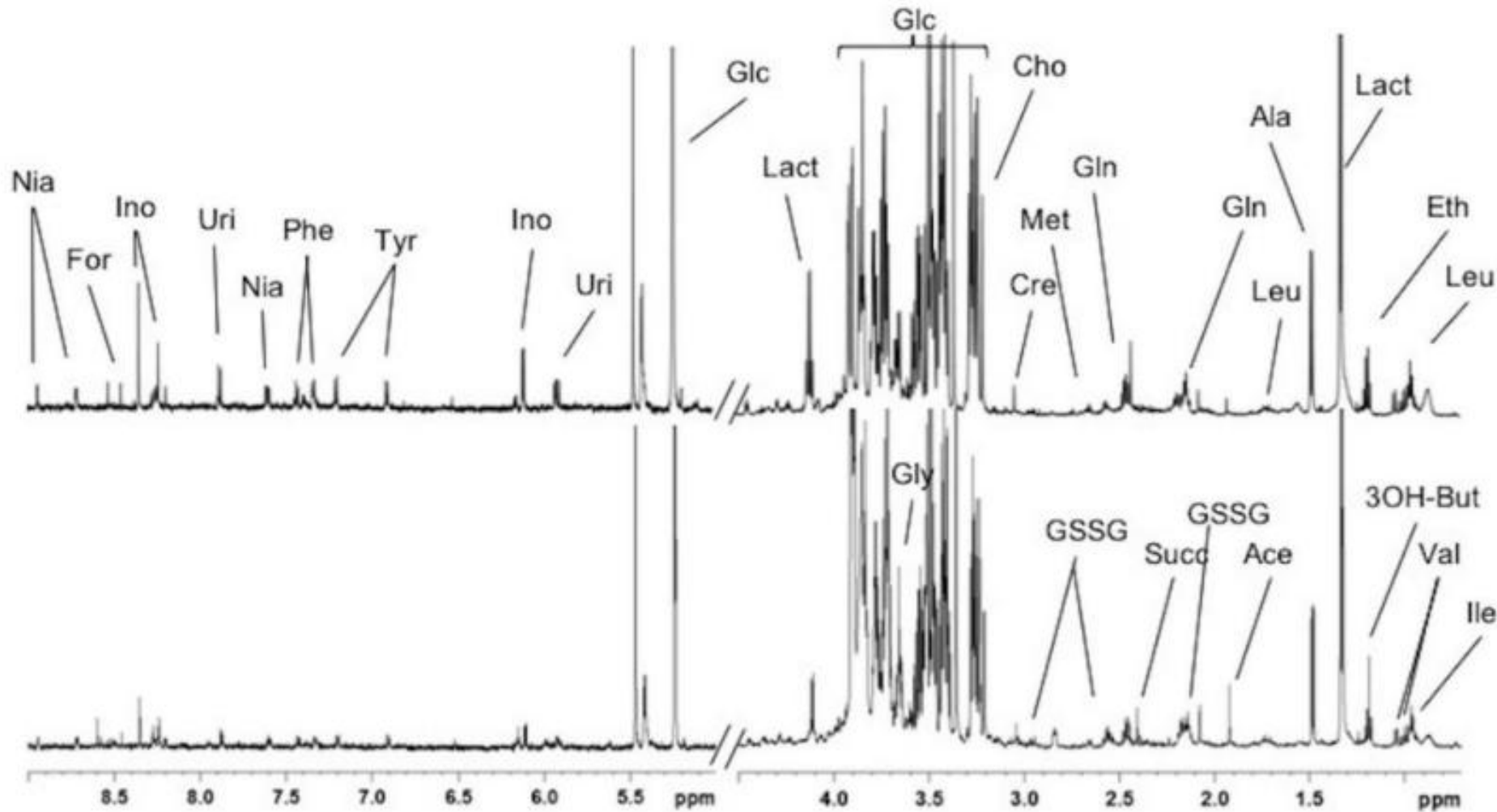
- Metabo/Lipido
- MS/NMR
- Transcrito/Lipido ...

Technology and compromises

❖ NMR vs MS?



For polar metabolites



❖ NMR

- Profiling
- Quantification
- Robustness

Figure 2. 600 MHz ¹H-NMR spectra of aqueous extracts from mouse liver samples. The Folch extraction

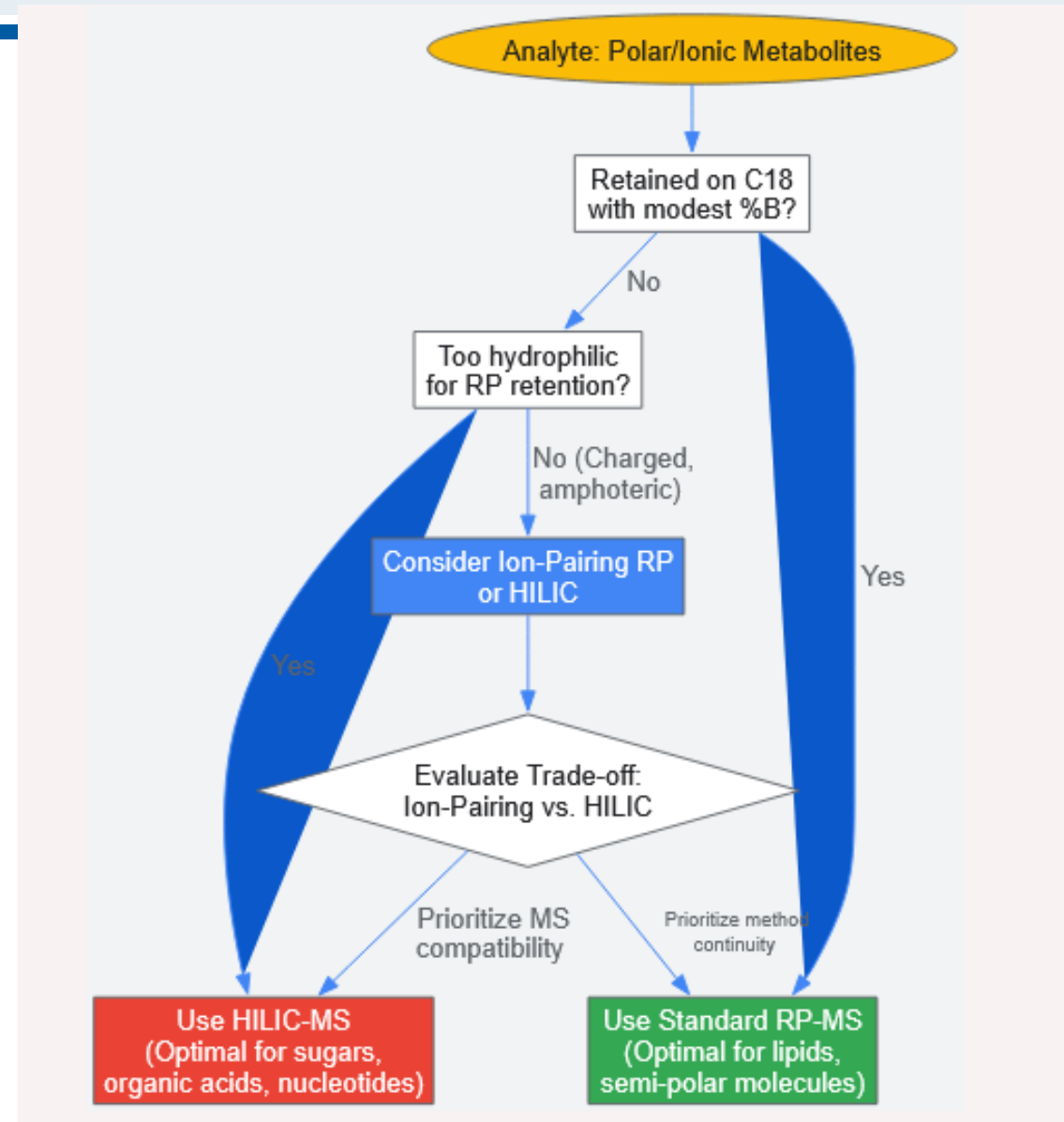
Amiel, Metabolites, 2020

For polar metabolites

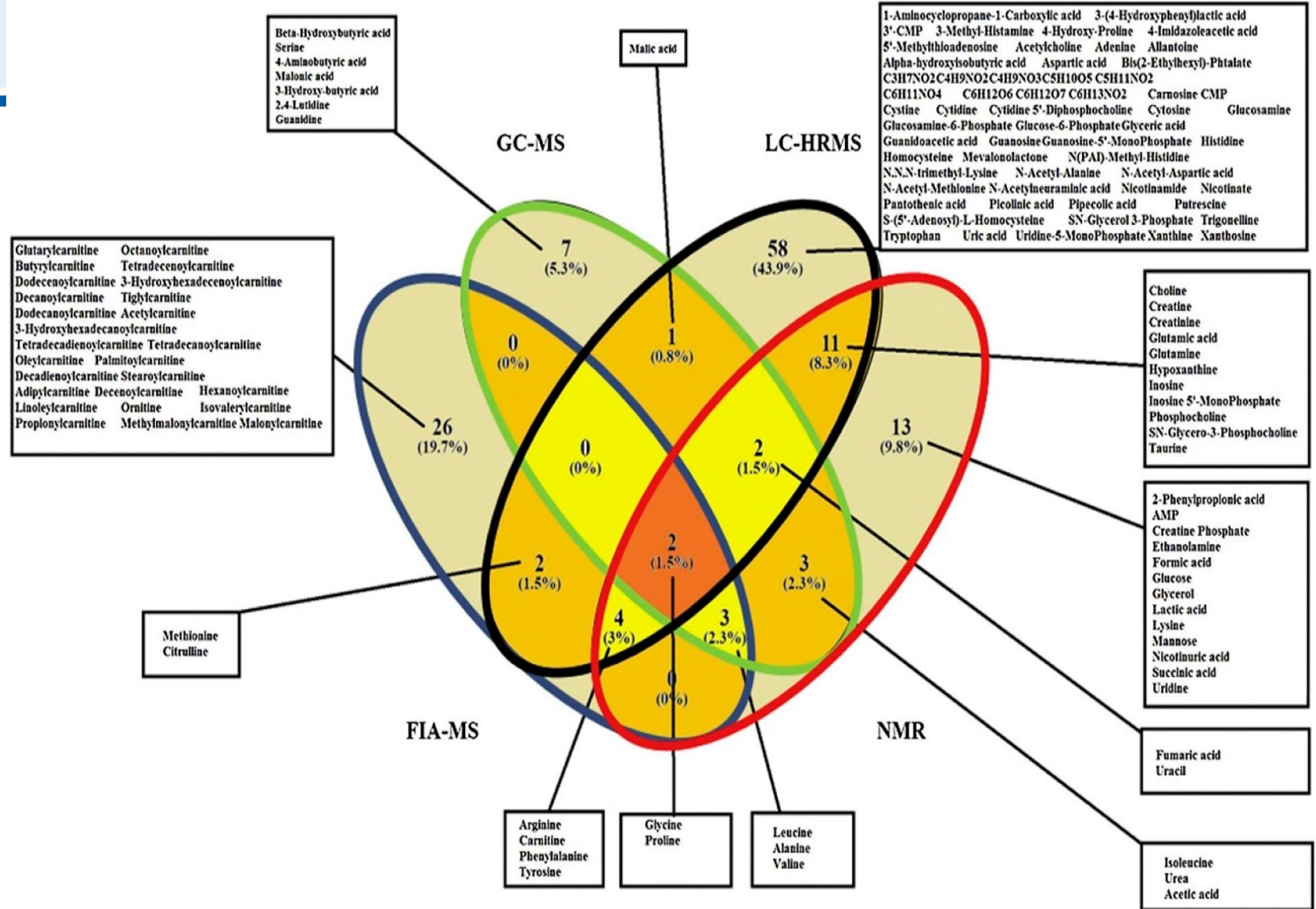
❖ HILIC or RP for polar metabolites ?

❖ GC for more volatile metabolites

- Dérivatisation
- Volatilome



For polar metabolites



Apolar metabolites

❖ NMR of the organic phase

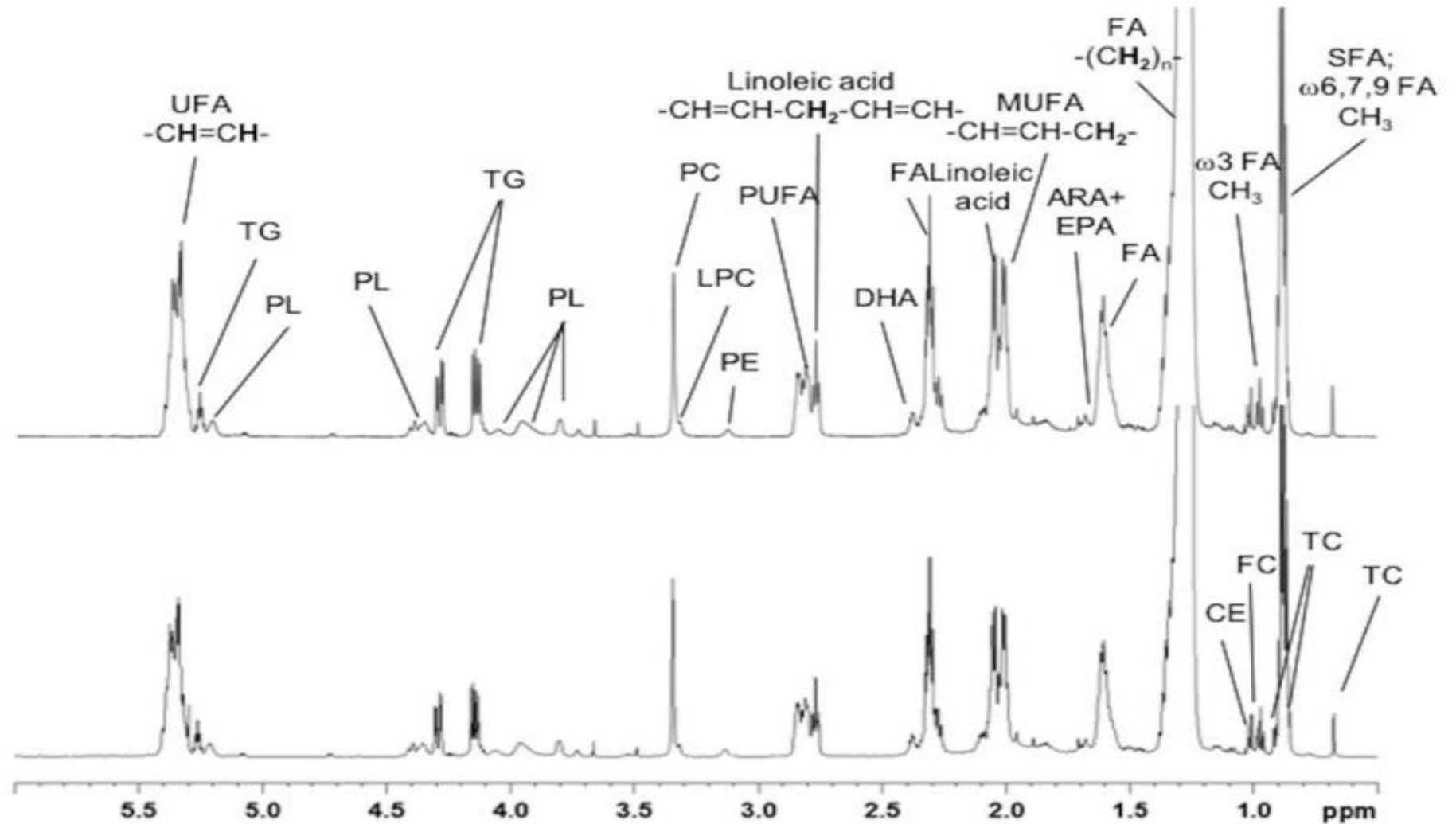
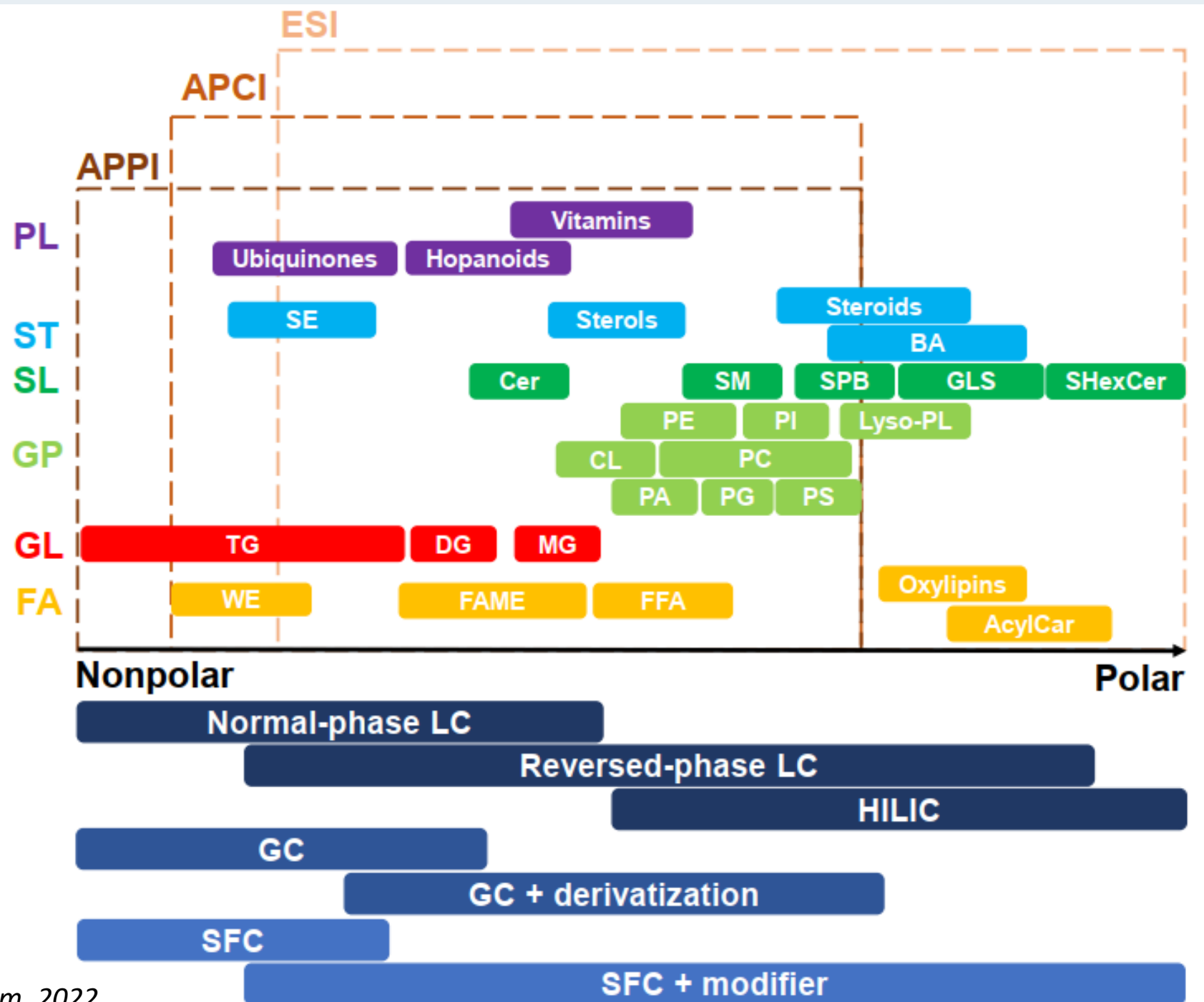


Figure 1. 600 MHz ^1H -NMR spectra of lipophilic extracts from mouse liver samples. The Folch extraction

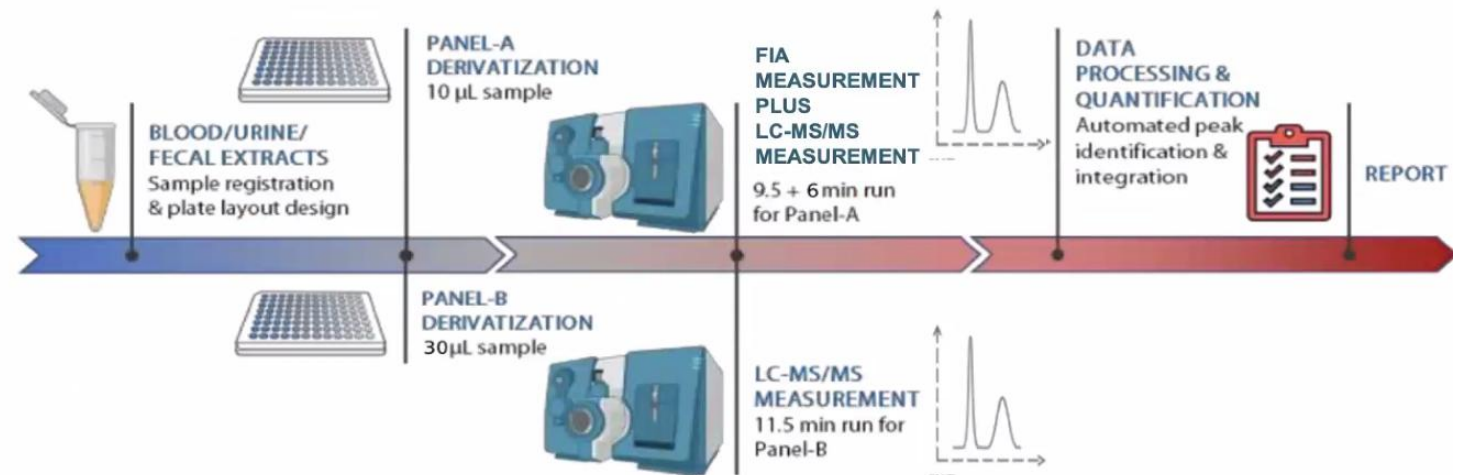
Apolar metabolites

❖ Importance of chromatography

❖ FIA & Shotgun



Kit technology

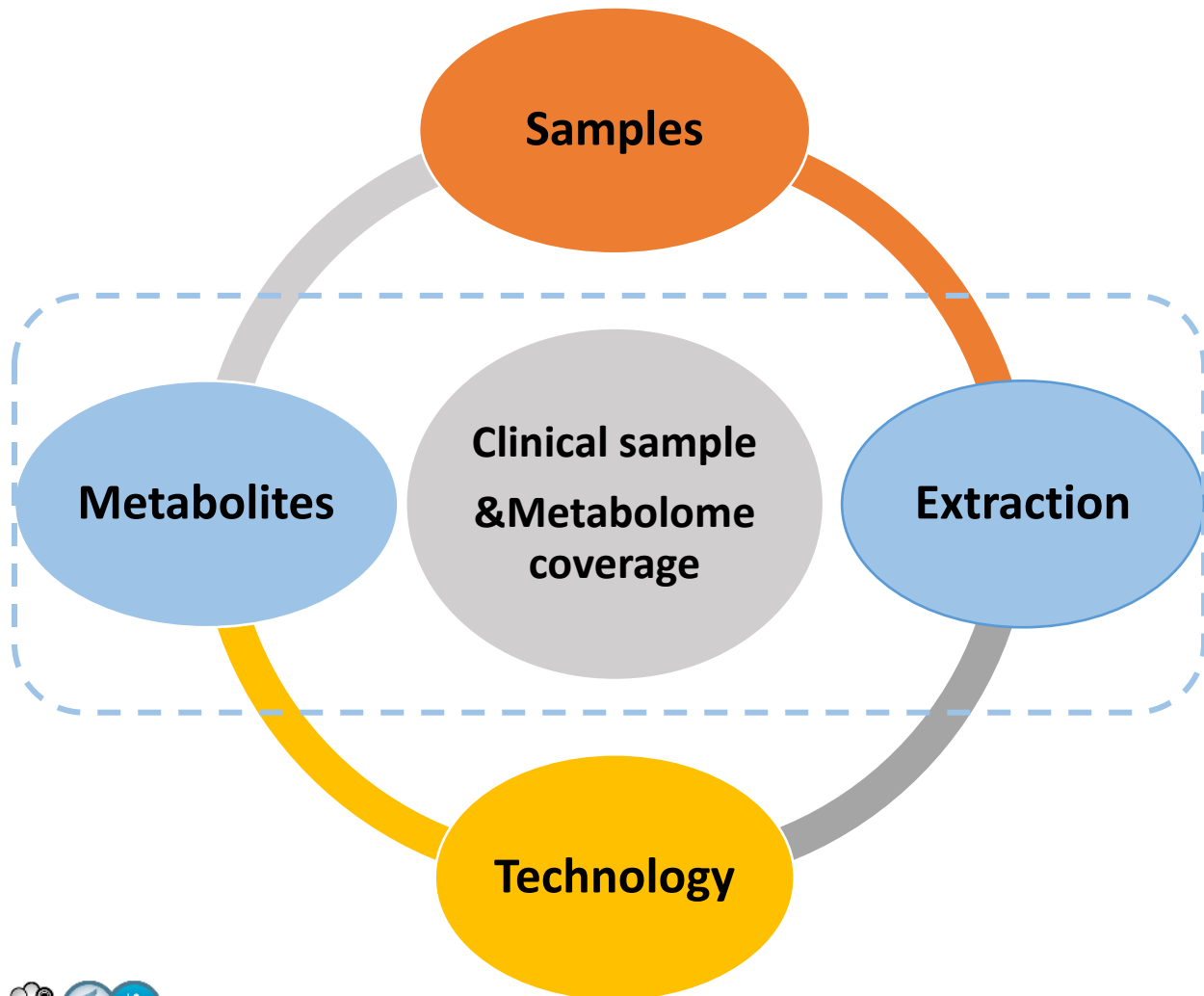


❖ Limited metabolome coverage

❖ High robustness

❖ Quantitative

To conclude



❖ The biological question is central

❖ Discovery or diagnostic?

❖ Available type of sample

❖ ...

Thanks

